& Davidson, 2023; Pasch & Cutura, 2024), it still exhibits notable limitations—particularly in nuanced or context-dependent classification tasks (Mu et al., 2023).

3.3. Training Transformer Model for LLM-Refusal

A common approach for training models for domain- or task-specific text classification is to fine-tune transformer-based models, such as BERT (Devlin et al., 2019) or RoBERTa (Liu et al., 2019). These models are pre-trained on large corpora of general text but fine-tuning them on a specific task allows them to better capture the nuances and context of the target domain. Fine-tuning requires training the model on task-specific labeled data while using transfer learning to retain the general language understanding from the pre-trained weights. This approach is widely used for classification tasks in NLP and, such as sentiment analysis (González-Carvajal & Garrido-Merchán, 2020), culture classification (Koch & Pasch 2023), or sustainability analyses in finance (Pasch & Ehnes 2022).

To train a model for our LLM-Refusal classification task, we used the labeled dataset of 1,750 prompts and 3,500 model responses, split into training, validation, and test sets following a 70-10-20 split, a widely adopted practice in supervised machine learning. The split was conducted at the prompt/question level, ensuring that both responses from the two models corresponding to the same prompt were allocated to the same set.

For the classification task, we combined the prompt and model response into a single input sequence, separated by distinct markers. To accommodate the 512-token limit imposed by BERT and RoBERTa, we truncated the prompt to the first 200 characters, ensuring that a sufficient portion of the response was always included in the input. This approach maintains the essential context of the prompt while prioritizing the model's ability to process the full response.

To ensure consistent and reliable performance, we adopted a standardized training configuration for the transformer-based model. Our approach followed established best practices for large-scale transformer training, focusing on stability, convergence, and handling class imbalance (Sun et al., 2019). The model was trained for 12 epochs with a learning rate of 1e-5, a batch size of 8, and weight decay of 0.01 to prevent overfitting. Training was further stabilized using the AdamW optimizer, known for its ability to combine adaptive learning rates with weight decay (Zheng et al., 2020).

Model evaluation was conducted at regular intervals, with validation every 3 epochs. The best checkpoint was selected based on the highest weighted F1 score, ensuring an optimal balance between precision and recall.

Table 3 summarizes the results on the hold-out test set on this text classification task. We compare the results of BERT and RoBERTa with a common standard approach of vectorizing texts as TF-IDF (term frequency-inverse document frequency) and then apply traditional supervised ML algorithms, such as logistic regression or random forest. (González-Carvajal &

Garrido-Merchán, 2020). We find that both BERT and RoBERTa outperform all benchmarks with RoBERTa slightly outperforming BERT. Despite the complexity of the multi-class classification task involving 5 distinct categories, RoBERTa demonstrates strong performance with an F1 score and accuracy of 88%. Achieving this level of performance in a multi-class setting is noteworthy, especially when compared to a random baseline of 20% accuracy for a 5-class problem. This result highlights RoBERTa's ability to effectively distinguish between the nuanced categories for LLM-response moderation.

Table 3: Text Classification on LLM-Refusal

Model	F1	Accuracy
Random Classifier	0.20	0.20
Tf-IDf + Supervised ML		
Logistic Regression	0.70	0.72
Random forest	0.78	0.80
XGBoost	0.78	0.78
Fine-Tuned Transformer		
BERT	0.87	0.87
RoBERTa	0.88	0.88

Table 3: Model performance overview. *F1* is the weighted average f1 score. Accuracy is the share of predictions in the test-set classified correctly.

We use this fine-tuned RoBERTa model to classify the response behavior of all model responses in the Chatbot Arena dataset. Table 4 presents the distribution of predicted categories based on this classification. As expected, the majority of responses are classified as "Normal" (87%), reflecting the broader nature of the dataset compared to the hand-labeled subset, which was pre-filtered to focus on responses likely to involve refusals or disclaimers.

3.4. Measuring Refusal Framing

Beyond classifying responses from the Chatbot Arena dataset based on their refusal and disclaimer behavior, we also extract additional semantic and linguistic features from the model outputs. This allows us to examine not only whether a refusal occurs, but also how it is delivered testing hypotheses H3a and H3b. These hypotheses focus on the idea that user satisfaction is not just affected by the presence of a refusal, but also by its formulation—whether it aligns with the user's query and whether it exhibits sufficient conversational effort. To capture these dimensions, we include two framing-related features: textual similarity and response length.

Textual Similarity: To proxy conversational alignment, we measure the semantic similarity between the user prompt and the model's response. Textual similarity serves as an indicator of alignment because it reflects how much the model engages with the specific content and intent of the user's request. In the context of refusals, this metric helps assess whether the model meaningfully acknowledges and responds to the prompt, even when it declines to fulfill it.

Formulaic refusals typically receive low similarity scores, as they avoid referencing the substance of the user's request. In contrast, more tailored refusals—those that explain the limitation in direct relation to the user's prompt—tend to yield higher similarity scores, signaling a more attentive and user-aware interaction.

To compute similarity, we use the all-MiniLM-L6-v2 sentence-transformer model (Reimers & Gurevych, 2019), which encodes both the prompt and the response into vector embeddings and calculates their cosine similarity. Higher scores indicate a greater degree of semantic overlap, and thus stronger conversational alignment.

Response Length: In addition to textual similarity, we also consider the length of the model's response, measured as the number of characters in the output. Longer responses are generally associated with higher perceived effort, thoroughness, and social presence—all of which may contribute to a more favorable user impression, even in the context of refusal.

Generic refusals (e.g., "*I'm sorry, I can't help with that*") are typically short and lacking in elaboration. More extended refusals, by contrast, often include contextualization, justification, or alternative suggestions—features that may mitigate the negative impact of the refusal by signaling helpfulness or empathy (Huang et al., 2024).

While many formulaic refusals tend to be both short and low in similarity, the two features do not always coincide. A response may be long yet only loosely connected to the user's prompt—for instance, by elaborating on an unrelated alternative rather than addressing the original request. Conversely, a refusal may be brief but semantically aligned, directly referencing the topic or intent of the prompt in a concise form. Together, these variables serve as complementary indicators of how refusals are framed and interpreted.

For easier interpretation, both the textual similarity and text length are standardized as Z-scores.

3.5. Content Sensitivity via Moderation API

To test H4, we examine whether user responses to ethical refusals differ depending on the sensitivity of the underlying prompt. Specifically, we ask whether refusals are penalized less when they respond to clearly sensitive prompts—such as those involving illegal, violent, or harmful content.

To quantify prompt sensitivity, we apply the OpenAI Moderation API (OpenAI, 2024b) to each prompt that elicited an ethical refusal. The OpenAI Moderation API has also been used in recent research to assess harmful content and evaluate LLM alignment (e.g., Franco et al., 2024; Chiang et al., 2024), supporting its relevance for studying safety-motivated behaviors in model outputs. The API detects potential policy violations across several categories, including *violence*, *self-harm*, *hate*, *sexual content*, and *illegal activities*. It returns both binary flags and category-specific classifications, enabling us to distinguish between refusals triggered by flagged content and those issued in response to unflagged prompts.